

Blur Detection Batch Processing System

Non-Technical / Quality Inspector User Manual

Ubuntu Desktop Edition

This guide is for daily operators and **does not require programming knowledge**. The Ubuntu computer, AI models, and desktop launcher must be prepared by IT before the first use.

I. Before You Start

1. Prepare the computer and photos

- Log in to the Ubuntu desktop and confirm that you can see the **Blur Detection** icon.
- If the photos are on a portable hard drive, USB drive, or portable SSD, connect it and open it once in the Files application.
- Make sure you know which folder contains the photos and where the results should be saved.
- If the desktop icon is missing or this computer has never run the system, contact IT.

Operators do not use commands: IT prepares the application, Conda environment, model files, and desktop launcher. For normal work, use the desktop icon and web page only.

2. Check the image names — this is essential

Every filename must use **SerialNumber-Position**, with one hyphen between the serial number and position.

Example	Correct?	Reason
1359-3.jpg	Correct	Serial number 1359, position 3.
4582A-9.bmp	Correct	Serial number 4582A, position 9.
image_3.jpg	Incorrect	It does not follow the required hyphen format.
1359_pos3.png	Incorrect	Extra text appears before the position.

Each product normally has images for positions **1, 3, 5, 7, and 9**. Incorrect names or missing positions will be sent for review.

Most important: A Terminal window opens with the application. Keep it open while processing and reviewing. Closing it stops the system and makes the web page unavailable.

II. Start the Ubuntu Application

- 1 Connect the photo drive**
If using a portable drive, connect it and open it in Files so Ubuntu can access it.
- 2 Open Blur Detection**
Double-click the **Blur Detection** icon on the desktop. If Ubuntu shows an “Allow Launching” option, right-click the icon, choose “Allow Launching,” then double-click it again.
- 3 Wait for the checks**
The Terminal verifies the application and starts the service. Keep it open. The first start may take longer.
- 4 Open the web page**
The page normally opens automatically. If it does not, open Chrome or Firefox, enter `http://localhost:8501`, and press Enter.

How to copy an Ubuntu folder path

1. Open the folder in the Files application.
2. Press `Ctrl+L`. The folder location appears at the top.
3. Press `Ctrl+C` to copy it.
4. Return to the Blur Detection page and press `Ctrl+V` in the correct path box.

Folder type	Example path
Folder in your home directory	<code>/home/operator/Pictures/batch_01</code>
Folder on a portable drive	<code>/media/operator/DRIVE_NAME/batch_01</code>

- 5 Choose input and output folders**
On “Batch Process,” paste the photo folder into “Input Raw Image Folder Path.” Paste the folder where results should be saved into “Result Output Root Folder Path.” Do not use the same folder for both.
- 6 Start processing**
Confirm that the device is `cuda`, then click “Run Batch Pipeline Process.”

How do I know it is finished? The page displays “Batch Processing Execution Completed Successfully!” with totals and processing time. Until then, do not disconnect the drive or close the browser or Terminal.

III. Review and Annotate Images

After batch processing, select “Review & Annotate.” The prediction file and processed-image folder normally update automatically from the output folder you selected.

Review in this order

- **Choose records:** Use the status filter to show Review, FLC, or NG records. If unsure, leave all statuses selected.
- **Enlarge an image:** Hold the mouse pointer over an image. Move it away to return to normal.
- **Check every position:** If the system is wrong, click **O**, **F**, **SN**, **N**, or **NG** beside the image.
- **Confirm the product:** Use “Need FLC,” “Don’t Need FLC,” or “NG (Fail Directly).”
- **Save and continue:** Click “Next” or press →. Use “Previous” or ← to go back.

1. What do the five labels mean?

Label	Meaning	Use it when
O	OK / Normal	The target position is normal, with no clear offset or distance problem.
F	Far	The target is clearly too far away, severely offset, or incomplete.
SN	Slightly Near	The target is slightly too near, but its boundary can still be seen.
N	Near	The target is much too near, badly blurred or enlarged, or an important area is missing.
NG	Abnormal / Reject	An image is missing, processing failed, or there is an unacceptable serious problem.

2. When does the system automatically require FLC?

FLC means that further abnormality review is required. The system requires FLC if any of these occur across positions 1, 3, 5, 7, and 9:

- At least one position is **N**;
- At least one position is **NG**;
- Three or more positions are **SN**.

Operator tip: The three product-level buttons set the whole product status. If you change a position afterward, the system calculates the status again. Check all five positions before giving the final product status.

IV. View Results and Finish Work

1. Which files should an operator open?

Open the output folder you selected, then open `dataset_output`. For normal work, use the Excel files ending in `.xlsx`.

File	Purpose
<code>consolidated_batch_predictions.xlsx</code>	System results for every product.
<code>predictions_filtered.xlsx</code>	Only products that need human checking or FLC review.
<code>human_annotated_predictions.xlsx</code>	Final results saved during human review. This is the main file after inspection is complete.
<code>raw_predictions.xlsx</code>	Detailed results for individual images; normally used only for troubleshooting.

Do not change system files: Files ending in `.json` are used by the system. Do not delete, rename, or edit them. The `processed_images` folder contains processed pictures.

2. Finish work safely

1. Confirm that processing is finished and review changes are saved.
2. Click the Terminal window and press `Ctrl+C`.
3. Wait for the application to stop. The Terminal may close automatically.
4. Close the browser. If using a portable drive, safely eject it afterward.

3. What should I do if something goes wrong?

What you see	What the operator can do
The desktop icon does not start	Right-click it and look for "Allow Launching." If unavailable, contact IT.
The browser did not open	Keep the Terminal open and visit <code>http://localhost:8501</code> .
The Terminal shows an error and waits	Photograph the full error. Do not change commands or files; contact IT.
The page shows <code>cpu</code>	Stop the program and contact IT. Normal operation should show <code>cuda</code> .
Input folder does not exist	Open the folder in Files, press <code>Ctrl+L</code> , and copy the path again.
Cannot save to a portable drive	Confirm the drive is connected and writable. Choose a folder in your home directory or contact IT.
Many products show missing images	Check filenames and confirm positions 1, 3, 5, 7, and 9 exist for each serial number.

Still not working? Do not modify the program. Photograph the Terminal and web-page error, then send the photos, selected folder paths, and time of the problem to IT.